

Unit 5 - Singular Value Decomposition

Lecture 1 - MATLAB class number 9 - QR factorization

Lecture 2 - MATLAB class number 10 - Eigen Values & Eigen vectors

Lectures 3 & 4 - Quadratic forms, Definitions of positive definite, negative definite, positive semi-definite, negative semi-definite, Indefinite forms & matrices.

→ If A is a real symmetric matrix of order n by n , then there exists an orthonormal matrix V whose columns are the eigen vectors of A and a diagonal matrix D , having its diagonal entries as the eigen values of A , such that $A = VDV^T$

→ The above process gives the eigen value decomposition of the matrix A .

→ The Singular Value Decomposition (SVD) is intimately related to the eigen value decomposition.

Quadratic form

→ Any $n \times n$ real symmetric matrix A determines a quadratic form q_A in n variables by the formula:

$$\begin{aligned} q_A(x_1, \dots, x_n) &= \sum_{i=1}^n \sum_{j=1}^n a_{ij} \cdot x_i \cdot x_j \\ &= x^T \cdot A \cdot x \end{aligned}$$

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→ Conversely, given a quadratic form in 'n' variables, its coefficients can be arranged into an 'n × n' Symmetric matrix.

→ Note : $x^T \cdot A \cdot x$ is a scalar

$$\therefore x^T \cdot A \cdot x = (x_1 \ x_2 \ \dots \ x_m) \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1m} \\ a_{21} & a_{22} & \dots & a_{2m} \\ \vdots & & & \\ a_{m1} & a_{m2} & \dots & a_{mm} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{pmatrix}$$

$$\begin{aligned} &= x_1(a_{11} \cdot x_1 + a_{12} \cdot x_2 + \dots + a_{1m} \cdot x_m) + \\ &\quad x_2(a_{21} \cdot x_1 + a_{22} \cdot x_2 + \dots + a_{2m} \cdot x_m) + \dots \\ &\quad + x_m(a_{m1} \cdot x_1 + a_{m2} \cdot x_2 + \dots + a_{mm} \cdot x_m) \\ &= \sum_{i \leq j}^m a_{ij} \cdot x_i \cdot x_j \end{aligned}$$

Quadratic form - Example 1

Let $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$. compute $x^T \cdot A \cdot x$ for the

following matrices :-

a) $A = \begin{bmatrix} 4 & 0 \\ 0 & 3 \end{bmatrix}$ b) $A = \begin{bmatrix} 3 & -2 \\ -2 & 7 \end{bmatrix}$

Solution :- a) $x^T \cdot A \cdot x$

$$= \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 4 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$= \cancel{4x_1/3} \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 4x_1 \\ 3x_2 \end{bmatrix}$$

$$a) \quad x^T \cdot A \cdot x = 4x_1^2 + 3x_2^2$$

$$b) \quad x^T \cdot A \cdot x = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 3 & -2 \\ -2 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$= \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 3x_1 - 2x_2 \\ -2x_1 + 7x_2 \end{bmatrix}$$

$$= x_1(3x_1 - 2x_2) + x_2(-2x_1 + 7x_2)$$

$$= 3x_1^2 - 2x_1x_2 - 2x_1x_2 + 7x_2^2$$

$$\boxed{x^T \cdot A \cdot x = 3x_1^2 - 4x_1x_2 + 7x_2^2}$$

→ If A isn't symmetric, the function $Q(x) = x^T A x$ is still a quadratic form :-

$$\text{Define } \hat{A} = \frac{A + A^T}{2}$$

$$x^T \cdot \hat{A} \cdot x = x^T \left\{ \frac{A + A^T}{2} \right\} x$$

$$= x^T \left\{ \frac{A}{2} + \frac{A^T}{2} \right\} x$$

$$= \frac{1}{2} x^T A x + \frac{1}{2} x^T A^T x$$

$$= \frac{1}{2} \{ x^T \cdot A \cdot x + x^T \cdot A \cdot x \}$$

$$= x^T \cdot A \cdot x$$

∴ Because of this, it is safe to assume that A is symmetric when we examine quadratic forms.

→ The quadratic form of $f(x, y) = ax^2 + 2abx + by^2$ may be represented as $x^T A x$ as :-

$$x = \begin{bmatrix} x \\ y \end{bmatrix} \quad A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

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Quadratic form : classification :-

→ Based on the sign of the quadratic forms, SVD can be classified into 3 categories :-

i) Positive definite if quadratic form > 0

ii) Positive semidefinite if quadratic form ≥ 0

iii) Negative definite if quadratic form < 0 .

$A^T A$ is positive definite.

$A^T A$ is symmetric & square

$$\text{Quadratic form: } x^T \cdot A^T \cdot A \cdot x = (Ax)^T \cdot (Ax) \\ = \|Ax\|^2 > 0$$

Example $A = \begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}$; $A^T A = \begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix} = \begin{bmatrix} 10 & -1 \\ -1 & 5 \end{bmatrix}$

Positive definiteness of $A+B$:-

→ If A & B are positive definite, then so is $A+B$.

$$\rightarrow x^T(A+B)x = x^T A x + x^T B x > 0 \quad \because x^T A x > 0 \text{ \& } x^T B x > 0$$

Positivity of Eigen values :-

Every Eigen value of a positive definite matrix is positive.

Proof :- Suppose A be a positive definite matrix.

Let λ be an eigen value of A and s be an eigen vector of A corresponding to λ . We have

$$As = \lambda s$$

It follows that :- $S^T A S = \lambda(S^T S)$

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$$\text{Hence } \lambda = \frac{S^T A S}{S^T S} > 0$$

A matrix is positive definite if every eigen value of the matrix is positive.

Proof :- Suppose every eigen value of A is positive.

By spectral theorem, A has an eigen value decomposition

$A = Q \Lambda Q^T$. It follows that

$$x^T A x = \underbrace{x^T}_{y^T} \cdot Q \Lambda Q^T \cdot \underbrace{x}_y = y^T \Lambda y = \sum_i \lambda_i y_i^2$$

Hence, the quadratic form $x^T A x$ is positive for any $x \neq 0$ and A is positive definite.

There are many ways to say a matrix is positive definite :-

- 1) A is positive definite
- 2) Every eigen value of A is positive
- 3) The determinant of every leading principal sub matrices of A is positive.
- 4) A has full positive pivots.

Example :- $A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$ The quadratic form of A is

The quadratic form of A is $x^T A x$

$$\begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

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$$\begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$= \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} \begin{bmatrix} 2x_1 - x_2 \\ -x_1 + 2x_2 - x_3 \\ -x_2 + 2x_3 \end{bmatrix}$$

$$\begin{aligned} x^T A x &= x_1(2x_1 - x_2) + x_2(-x_1 + 2x_2 - x_3) + x_3(-x_2 + 2x_3) \\ &= 2x_1^2 - x_1x_2 - x_1x_2 + 2x_2^2 - x_2x_3 - x_2x_3 + 2x_3^2 \end{aligned}$$

$$x^T A x = 2x_1^2 + 2x_2^2 + 2x_3^2 - 2x_1x_2 - 2x_2x_3$$

$$= 2(x_1^2 - x_1x_2) + 2(x_2^2 + x_3^2 - x_2x_3) + x_2^2$$

$$= 2x_1^2 + x_2^2 - x_1x_2 - x_1x_2 + x_1x_2$$

Example 0- For what numbers 'b' is the following matrix semidefinite?

$$A = \begin{bmatrix} 2 & -1 & b \\ -1 & 2 & -1 \\ b & -1 & 2 \end{bmatrix}$$

Solution 0- $\begin{vmatrix} 2 & -1 & b \\ -1 & 2 & -1 \\ b & -1 & 2 \end{vmatrix} = 2(4-1) + 1(-2+b) + b(1-2b) \geq 0$

$$2(3) + -2 + b + b - 2b^2 \geq 0$$

$$6 - 2 + 2b - 2b^2 \geq 0$$

$$-2b^2 + 2b + 4 \geq 0$$

$$b^2 - b - 2 \leq 0$$

$$b^2 - 2b + b - 2 \leq 0$$

$$b(b-2) + 1(b-2) \leq 0$$

$$(b-1)(b-2) \leq 0$$

$$1 \leq b \leq 2$$

If A is a symmetric matrix, then it is said to be positive semidefinite if :-

i) $x^T A x \geq 0 \quad \forall x$

ii) All eigen values of A are greater than or equal to zero. i.e. $\lambda \geq 0$

iii) None of the principal submatrices have negative determinants.

iv) Pivots are non negative

v) There exists a matrix R , possibly with dependent columns such that $A = R^T \cdot R$.

Example :- Write the symmetric matrix which corresponds to the following QF ?

a) $Q(x) = 5x_1^2 + 3x_2^2 + 2x_3^2 - x_1x_2 + 8x_2x_3$

Solution :- The coefficients of x_1^2, x_2^2, x_3^2 go on to the diagonal of A .

To make A symmetric, the coefficients of $x_i x_j$ for $i \neq j$ must be split evenly between (i, j) and (j, i) positions in A .

oo $A = \begin{bmatrix} 5 & -1/2 & 2 \\ -1/2 & 3 & 4 \\ 0 & 4 & 2 \end{bmatrix}$

b) $5x_1^2 - x_2^2 + 7x_3^2 + 5x_1x_2 - 3x_1x_3$

$$A = \begin{bmatrix} 5 & 5/2 & -3/2 \\ 5/2 & -1 & 0 \\ -3/2 & 0 & 7 \end{bmatrix}$$

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$$c) \quad x_3^2 - 4x_1x_2 + 4x_2x_3$$

Solution :-

$$\begin{bmatrix} 0 & -2 & 0 \\ -2 & 0 & 2 \\ 0 & 2 & 1 \end{bmatrix}$$

Lectures 5 to 7 - Tests for Positive Definiteness, Problems on Positive Definite Matrices & Least Squares, Problems on Semi-definite Matrices :-

Tests for Positive Definiteness :-

Example :- Let $A = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$ is a symmetric matrix

A is positive definite if $a > 0$ and $ac - b^2 > 0$ ($|A| > 0$)
 $\Rightarrow$ Both the eigen values are positive. Product of eigen values $\lambda_1 \cdot \lambda_2 > 0$.

$$Q(x) = x^T A x$$

$$= \begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} a & b \\ b & c \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$= \begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} ax + by \\ bx + cy \end{bmatrix}$$

$$= x(ax + by) + y(bx + cy)$$

$$= ax^2 + bxy + bxy + cy^2$$

$$= ax^2 + 2bxy + cy^2$$

$$= a\left(x^2 + 2\frac{b}{a}xy\right) + cy^2$$

$$= a\left\{x^2 + 2(x)\left(\frac{b}{a}y\right) + \frac{b^2y^2}{a^2} - \frac{b^2y^2}{a^2}\right\} + cy^2$$

$$= a\left(x + \frac{b}{a}y\right)^2 - \frac{ab^2y^2}{a^2} + cy^2$$

$$Q(x) = a \left(x + \frac{by}{a} \right)^2 - \frac{ab^2y^2}{a^2} + cy^2$$

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$$= a \left(x + \frac{by}{a} \right)^2 + cy^2 - \frac{b^2y^2}{a}$$

$$= a \left(x + \frac{by}{a} \right)^2 + \left(\frac{ac - b^2}{a} \right) y^2$$

Note: $a, \frac{ac - b^2}{a}$ are the pivots of the matrix A .

$x^T A x$ stays positive when n independent squares are multiplied by "positive pivots".

upper left submatrices of a matrix :-

consider $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$

The upper left submatrices are :-

$$A_1 = a_{11} ; A_2 = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} ; A_3 = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

If A is of order ' $n \times n$ ', then we can find ' n ' upper left submatrices of A .

Tests for positive definiteness of a matrix A :-

- 1) $x^T A x > 0$ for all non zero real vectors x .
- 2) All the eigen values are positive.
- 3) All the upper left submatrices have positive determinants.

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4) All the pivots (without row exchanges) satisfy $d_k > 0$ (are positive).

Example 0- Test for positive definiteness of the following matrices 0-

a) $A = \begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix}$

Solution 0- Test 1 $x^T A x > 0 \quad \forall x \neq 0$

If $A = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$ then $x^T A x = ax^2 + 2bxy + cy^2$

$= \begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix}$

$x^T A x = 2x^2 + 6xy + y^2$

$x^T A x = a \left(x + \frac{by}{a} \right)^2 + \left(\frac{ac - b^2}{a} \right) y^2$

$= 2 \left(x + \frac{3y}{2} \right)^2 + \left(\frac{10 - 9}{2} \right) y^2$

$x^T A x = 2 \left(x + \frac{3y}{2} \right)^2 + \frac{1}{2} y^2$

$\Rightarrow x^T A x > 0$

Test 2 Eigen values of A 0-

$A = \begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix}$

$|A - \lambda I| = 0 \Rightarrow \begin{vmatrix} 2-\lambda & 3 \\ 3 & 5-\lambda \end{vmatrix} = 0$

$(2-\lambda)(5-\lambda) - 9 = 0$

$10 - 2\lambda - 5\lambda + \lambda^2 - 9 = 0$

$\lambda^2 - 7\lambda + 1 = 0$

$\frac{7 \pm \sqrt{49 - 4(1)(1)}}{2} = \frac{7 \pm \sqrt{45}}{2}$

$\lambda_1 = \frac{7 + \sqrt{45}}{2} ; \lambda_2 = \frac{7 - \sqrt{45}}{2}$

$\lambda_1, \lambda_2 > 0 \Rightarrow A$ is positive definite

III upper left submatrices :-

$$A_1 = 2; A_2 = \begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix}$$

$$|A_1| = 2; |A_2| = 10 - 9 = 1$$

All the upper left submatrices are having positive determinants.

IV $\begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix} \xrightarrow{R_2 - \frac{3}{2}R_1} \begin{bmatrix} 2 & 3 \\ 0 & \frac{1}{2} \end{bmatrix}$

$$5 - \frac{3}{2}(3) = 5 - \frac{9}{2}$$

All pivots are positive.

By all the above 4 tests, gives A is positive definite.

2) $A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$ 3) $\begin{bmatrix} 1 & 3 \\ 3 & 5 \end{bmatrix}$

consider $\begin{bmatrix} 1 & 3 \\ 3 & 5 \end{bmatrix} = \begin{bmatrix} a & b \\ b & c \end{bmatrix} \xrightarrow{R_2 - 3R_1} \begin{bmatrix} 1 & 3 \\ 0 & -4 \end{bmatrix}$

we have a negative pivot. ∴ this matrix is not positive definite.

consider $\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$ $|A - \lambda I| = 0$

$$\begin{vmatrix} 2-\lambda & -1 & 0 \\ -1 & 2-\lambda & -1 \\ 0 & -1 & 2-\lambda \end{vmatrix} = 0$$

$$(2-\lambda)\{(2-\lambda)^2 - 1\} + 1\{1\} = 0$$

$$(2-\lambda)\{\lambda^2 + 4 - 4\lambda - 1\} + 1 = 0$$

$$(2-\lambda)\{\lambda^2 - 4\lambda + 3\} + 1 = 0$$

$$2\lambda^2 - 8\lambda^2 + 6 - \lambda^3 + 4\lambda^2 - 3\lambda + 1 = 0$$

$$-\lambda^3 - 2\lambda^2 - 3\lambda + 7 = 0$$

$$\lambda^3 + 2\lambda^2 + 3\lambda - 7 = 0$$

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$$\begin{vmatrix} 2-\lambda & -1 & 0 \\ -1 & 2-\lambda & -1 \\ 0 & -1 & 2-\lambda \end{vmatrix} = 0 \Rightarrow (2-\lambda)\{(2-\lambda)^2 - 1\} + 1\{-1(2-\lambda)\} = 0$$

$$(2-\lambda)\{\lambda^2 + 4 - 4\lambda - 1\} + (-2 + \lambda) = 0$$

$$(2-\lambda)\{\lambda^2 - 4\lambda + 3\} + \lambda - 2 = 0$$

$$2\lambda^2 - 8\lambda + 6 - \lambda^3 + 4\lambda^2 - 3\lambda + \lambda - 2 = 0$$

$$-\lambda^3 + 6\lambda^2 - 10\lambda + 4 = 0$$

$$\lambda^3 - 6\lambda^2 + 10\lambda - 4 = 0$$

$$2^3 - 6(4) + 10(2) - 4 = 0$$

$$8 - 24 + 20 - 4 = 0$$

$$0 = 0$$

$$\text{LHS} = \text{RHS}$$

∴ $\lambda = 2$ is an eigen value

$$\begin{array}{c|cccc} 2 & 1 & -6 & +10 & -4 \\ & 0 & 2 & -8 & 4 \\ \hline & 1 & -4 & 2 & 0 \end{array}$$

$$\lambda^2 - 4\lambda + 2 = 0$$

$$\lambda = \frac{4 \pm \sqrt{16 - 4(1)(2)}}{2}$$

$$\frac{4 \pm \sqrt{8}}{2} = \frac{4 \pm 2\sqrt{2}}{2} \Rightarrow 2 \pm \sqrt{2}$$

∴ Eigen values = 0, $2 + \sqrt{2}$, $2 - \sqrt{2}$

$2 \pm \sqrt{2} > 0 \Rightarrow$ Eigen values are positive.

Pivots ∴ $A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \xrightarrow{R_2 + 1/2 R_1} \begin{bmatrix} 2 & -1 & 0 \\ 0 & 3/2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$

$$0 - 2x = 0$$

$$0 = 2x$$

$$-1 - 2(x) = 0$$

$$-1 = 2x$$

$$x = -1/2$$

$$2^{-1/2}$$

$$\begin{bmatrix} 2 & -1 & 0 \\ 0 & 3/2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \xrightarrow{R_3 + 2/3 R_2} \begin{bmatrix} 2 & -1 & 0 \\ 0 & 3/2 & -1 \\ 0 & 0 & 4/3 \end{bmatrix}$$

$$-1 - 3/2 x = 0$$

$$-1 = 3/2 x \quad 2 = -2/3$$

$$x = -2/3$$

Here all the pivots are positive.

∴ A is positive definite.

Test for positive definiteness :-

$$1) \begin{bmatrix} -1 & 2 \\ 2 & -8 \end{bmatrix} \quad 2) \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & 1 \\ -1 & 1 & 2 \end{bmatrix} \quad 3) \begin{bmatrix} 3 & 2 & 0 \\ 2 & 2 & 2 \\ 0 & 2 & 1 \end{bmatrix} \quad 4) \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 1 \\ 2 & 1 & 0 \end{bmatrix}^2$$

Solution :- 1) $A = \begin{bmatrix} -1 & 2 \\ 2 & -8 \end{bmatrix}$

This is not positive definite as pivots are not positive.

$$2) \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & 1 \\ -1 & 1 & 2 \end{bmatrix}$$

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 2-\lambda & -1 & 1 \\ -1 & 2-\lambda & 1 \\ -1 & 1 & 2-\lambda \end{vmatrix} = 0$$

$$\begin{aligned} & 2\lambda^2 - 8\lambda + 6 - \lambda^2 + 4\lambda - 3 + \lambda^2 - 11\lambda + 6 = 0 \\ & = -\lambda^2 + 3 - 6\lambda^2 + 11\lambda - 6 = 0 \\ & = \lambda^2 - 4\lambda + 3 \end{aligned}$$

$$(2-\lambda)\{(2-\lambda)^2 - 1\} + 1\{-(2-\lambda) + 1\} + 1\{-1 + (2-\lambda)\} = 0$$

$$(2-\lambda)\{4 + \lambda^2 - 4\lambda - 1\} + \{-2 + \lambda + 1\} + \{-1 + 2 - \lambda\} = 0$$

$$(2-\lambda)\{\lambda^2 - 4\lambda + 3\} + (\lambda - 1) + 1 - \lambda = 0$$

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$$\lambda^3 - 6\lambda^2 + 11\lambda - 6 = 0$$

$$1 - 6 + 11 - 6 = 0$$

$$\lambda = 1 \left[\begin{array}{cccc} 1 & -6 & 11 & -6 \\ 0 & 1 & -5 & 6 \\ \hline 1 & -5 & 6 & 0 \end{array} \right]$$

$$\lambda^2 - 5\lambda + 6 = 0$$

$$\lambda^2 - 3\lambda - 2\lambda + 6 = 0$$

$$\lambda(\lambda - 3) - 2(\lambda - 3) = 0$$

$$(\lambda - 3)(\lambda - 2) = 0$$

$$\lambda = 2 \text{ \& } \lambda = 3$$

∴ eigen values are positive

$$A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & 1 \\ -1 & 1 & 2 \end{bmatrix} \quad A_1 = 2; A_2 = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}; A_3 = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & 1 \\ -1 & 1 & 2 \end{bmatrix}$$

$$\begin{aligned} |A_1| &= 2; |A_2| = 3; |A_3| = 2(4-1) + 1(-2+1) + 1(-1+2) \\ &= 2(3) + 1(-1) + 1 \\ &= 6 - 1 + 1 \\ &= 6 \end{aligned}$$

∴ All upper submatrices have positive determinants.

To find pivots:-

$$\begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & 1 \\ -1 & 1 & 2 \end{bmatrix} \xrightarrow[2R_2+R_1]{2R_2+R_1} \begin{bmatrix} 2 & -1 & 1 \\ 0 & 3 & 3 \\ 0 & 1 & 5 \end{bmatrix} \xrightarrow{3R_3-R_2} \begin{bmatrix} 2 & -1 & 1 \\ 0 & 3 & 3 \\ 0 & 0 & 12 \end{bmatrix}$$

All pivots are positive. ∴ A is positive definite.

$$3) \quad A = \begin{pmatrix} 3 & 2 & 0 \\ 2 & 2 & 2 \\ 0 & 2 & 1 \end{pmatrix}$$

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 3-\lambda & 2 & 0 \\ 2 & 2-\lambda & 2 \\ 0 & 2 & 1-\lambda \end{vmatrix} = 0$$

$$(3-\lambda)\{(2-\lambda)(1-\lambda)-4\}-2\{2(1-\lambda)\}=0$$

$$(3-\lambda)\{2-2\lambda-\lambda+\lambda^2-4\}-2\{2-2\lambda\}=0$$

$$(3-\lambda)\{\lambda^2-3\lambda-2\}-4+4\lambda=0$$

$$3\lambda^2-9\lambda-6-\lambda^3+3\lambda^2+2\lambda-4+4\lambda=0$$

$$-\lambda^3+6\lambda^2-3\lambda-10=0$$

$$\lambda^3-6\lambda^2+3\lambda+10=0$$

$$-1-6-3+10=0$$

$$\lambda = -1 \Rightarrow A \text{ is not positive definite}$$

$$|A_1| = 3; |A_2| = \begin{vmatrix} 3 & 2 \\ 2 & 2 \end{vmatrix} = 6-4=2$$

$$\begin{aligned} |A_3| &= 3(2-4)-2(2) \\ &= 3(-2)-4 \\ &= -10 < 0 \end{aligned}$$

$$4) \quad \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 1 \\ 2 & 1 & 0 \end{bmatrix}^2 = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 1 \\ 2 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 1 \\ 2 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 5 & 2 & 1 \\ 2 & 2 & 2 \\ 1 & 2 & 5 \end{bmatrix}$$

To find eigen values :- $|A - \lambda I| = 0$

$$\begin{vmatrix} 5-\lambda & 2 & 1 \\ 2 & 2-\lambda & 2 \\ 1 & 2 & 5-\lambda \end{vmatrix} = 0$$

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$$\begin{vmatrix} 5-\lambda & 2 & 1 \\ 2 & 2-\lambda & 2 \\ 1 & 2 & 5-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (5-\lambda)\{(2-\lambda)(5-\lambda)-4\} - 2\{2(5-\lambda)-2\} + 1\{4-(2-\lambda)\} = 0$$

$$(5-\lambda)\{10-2\lambda-5\lambda+\lambda^2-4\} - 2\{10-2\lambda-2\} + \{2+\lambda\} = 0$$

$$(5-\lambda)\{\lambda^2-7\lambda+6\} - 2\{8-2\lambda\} + \lambda+2 = 0$$

$$5\lambda^2-35\lambda+30-\lambda^3+7\lambda^2-6\lambda-16+4\lambda+\lambda+2=0$$

$$-\lambda^3+12\lambda^2-36\lambda+16=0$$

$$\lambda^3-12\lambda^2+36\lambda-16=0$$

$$64-192+144-16=0$$

$$208-208=0$$

$$\lambda=4$$

$$\lambda=4 \begin{vmatrix} 1 & -12 & 36 & -16 \\ 0 & 4 & -32 & 16 \\ 1 & -8 & 4 & 0 \end{vmatrix}$$

$$\lambda^2-8\lambda+4=0$$

$$\frac{8 \pm \sqrt{64-4(1)(4)}}{2}$$

$$\frac{8 \pm \sqrt{48}}{2} \Rightarrow \frac{8 \pm 4\sqrt{3}}{2} \Rightarrow 4 \pm 2\sqrt{3}$$

$$\lambda_1 > 0$$

Determinant test :-

$$A_1 = 5 > 0 ; A_2 = \begin{vmatrix} 5 & 2 \\ 2 & 2 \end{vmatrix} ; |A_2| = 6 > 0$$

$$A_3 = \begin{vmatrix} 5 & 2 & 1 \\ 2 & 2 & 2 \\ 1 & 2 & 5 \end{vmatrix} ; |A_3| = 5(10-4) - 2\{10-2\} + 1\{4-2\}$$

$$|A_3| = 5(6) - 2(8) + 1(2)$$

$$30 - 16 + 2$$

$$16 > 0$$

Pivot test :-

$$A = \begin{bmatrix} 5 & 2 & 1 \\ 2 & 2 & 2 \\ 1 & 2 & 5 \end{bmatrix} \xrightarrow[R_3 - \frac{1}{5}R_1]{R_2 - \frac{2}{5}R_1} \begin{bmatrix} 5 & 2 & 1 \\ 0 & 6/5 & 8/5 \\ 0 & 8/5 & 24/5 \end{bmatrix}$$

$$\begin{aligned} 2 - 5x &= 0 \\ 2 &= 5x \\ x &= 2/5 \end{aligned}$$

$$\frac{2 - \frac{4}{5}}{1} = \frac{2 - \frac{2}{5}}{1}$$

$$\begin{aligned} 1 - 5x &= 0 \\ 1 &= 5x \\ x &= 1/5 \end{aligned}$$

$$\begin{aligned} \frac{5 - \frac{1}{5}}{1} &= \frac{25 - 1}{5} \\ \frac{2 - \frac{2}{5}}{1} &= \frac{10 - 2}{5} \end{aligned}$$

$$\frac{8}{5} - \frac{6}{5}x = 0$$

$$\begin{aligned} \frac{8}{5} &= \frac{6x}{5} \\ x &= 8/6 \end{aligned}$$

$$\begin{bmatrix} 5 & 2 & 1 \\ 0 & 6/5 & 8/5 \\ 0 & 8/5 & 24/5 \end{bmatrix} \xrightarrow{R_3 - \frac{8}{6}R_2} \begin{bmatrix} 5 & 2 & 1 \\ 0 & 6/5 & 8/5 \\ 0 & 0 & 8/3 \end{bmatrix}$$

$$\frac{24}{5} - \frac{8}{3} \times \frac{8}{5} = \frac{24}{5} - \frac{32}{15} = \frac{72 - 32}{15} = \frac{40}{15} = \frac{8}{3}$$

All pivots are positive. ∴ A is positive definite

Example :- For which 's' & 't' do A & B have all $\lambda > 0$

$$A = \begin{bmatrix} s & -4 & -4 \\ -4 & s & -4 \\ -4 & -4 & s \end{bmatrix} \quad B = \begin{bmatrix} 1 & 2 & 4 \\ 2 & t & 8 \\ 4 & 8 & 7 \end{bmatrix}$$

Solution :- use eigen values test :-

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} s-\lambda & -4 & -4 \\ -4 & s-\lambda & -4 \\ -4 & -4 & s-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (s-\lambda) \left\{ (s-\lambda)^2 - 16 \right\} + 4 \left\{ -4(s-\lambda) - 16 \right\}$$

$$-4 \left\{ 16 + 4(s-\lambda) \right\} = 0$$

$$\Rightarrow (s-\lambda) \left\{ s^2 + \lambda^2 - 2s - 16 \right\}$$

$$+ 4 \left\{ -4s + 4s\lambda - 16 \right\}$$

$$-4 \left\{ 16 + 4s - 4\lambda \right\} = 0$$

$$B = \begin{bmatrix} 1 & 2 & 4 \\ 2 & t & 8 \\ 4 & 8 & 7 \end{bmatrix}$$

When B is positive definite?

Solution :- Eigen Value Test :-

$$|B - \lambda I| = 0$$

$$\begin{vmatrix} 1-\lambda & 2 & 4 \\ 2 & t-\lambda & 8 \\ 4 & 8 & 7-\lambda \end{vmatrix} = 0$$

$$\lambda^3 - (\sum d)\lambda^2 + (\sum m_d)\lambda - |A| = 0$$

$$\begin{aligned} \sum d &= 1-\lambda + t-\lambda + 7-\lambda \\ &= 1+t+7-3\lambda \end{aligned}$$

$$B_1 = 1 > 0 ; B_2 = \begin{bmatrix} 1 & 2 \\ 2 & t \end{bmatrix} \quad |B_2| = t-4$$

$$\begin{aligned} B_3 &= \begin{bmatrix} 1 & 2 & 4 \\ 2 & t & 8 \\ 4 & 8 & 7 \end{bmatrix} \quad |B| = 1(7t-64) - 2(14-32) \\ &\quad + 4(16-4t) \\ &= 7t-64-28+64+64-16t \\ &= -9t+36 \end{aligned}$$

for positive definiteness :- B_1, B_2, B_3 should be greater than 0.

$$\begin{aligned} (t-4) &> 0 & -9t+36 &> 0 \\ & & 9t-36 & > 0 \\ & & t-4 & > 0 \end{aligned}$$

∴ There is no value of t to satisfy $t > 4$ & $t < 4$. ∴ No value of t makes B positive definite.

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$$A = \begin{bmatrix} S & -4 & -4 \\ -4 & S & -4 \\ -4 & -4 & S \end{bmatrix}$$

$$A_1 = S ; |A_1| = S > 0$$

$$A_2 = \begin{bmatrix} S & -4 \\ -4 & S \end{bmatrix} ; |A_2| = S^2 - 16 > 0$$

$$S^2 > 16$$

$$S > 4$$

$$A_3 = \begin{bmatrix} S & -4 & -4 \\ -4 & S & -4 \\ -4 & -4 & S \end{bmatrix}$$

$$|A_3| = S \{ S^2 - 16 \} + 4 \{ -4S - 16 \} - 4 \{ 16 + 4S \} \Rightarrow$$

$$= S^3 - 16S - 16S - 64 - 64 - 16S$$

$$= S^3 - 48S - 128$$

$$= -8^3 - 48(-8) - 128$$

$$= -512 + 384 - 128$$

Positive Definite Matrices and Least Squares :-

→ Consider the least squares problem in the form $Rx = b$.

→ Here R has m equations with $m > n$.

→ The least-square solution $\hat{x}$ is of the form $R^T R \hat{x} = R^T b$.

→ Here $R^T R$ is not only symmetric but also it is positive provided ' n ' columns of R are linearly independent.

"The symmetric matrix A is positive definite iff there is a matrix ' R ' with independent columns such that $A = R^T \cdot R$ ".

$$\begin{aligned} x^T A x &= x^T \{ R^T R \} x \\ &= x^T \cdot R^T \cdot R x \\ &= (R x)^T \cdot R x \\ &= \| R x \|^2 > 0 \end{aligned}$$

$$x^T A x > 0 \Rightarrow x^T (R^T R) x > 0$$

$\Rightarrow R^T R = A$ is positive definite.

To find R , we use elimination :-

$$\begin{aligned} A &= L D L^T \\ &= L \sqrt{D} \sqrt{D} L^T \\ &= (L \sqrt{D}) (\sqrt{D} L^T) \end{aligned}$$

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choose $R = \sqrt{D} L^T$

$$R^T = (\sqrt{D} L^T)^T = (L^T)^T (\sqrt{D})^T \\ = L \sqrt{D}$$

$$A = R^T R \quad \text{where } R = \sqrt{D} \cdot L^T$$

This is called Cholesky decomposition

To find R , we use eigen values

$$A = Q \Lambda Q^T \\ = Q \sqrt{\Lambda} \cdot \sqrt{\Lambda} \cdot Q^T \\ = Q \sqrt{\Lambda} \cdot \sqrt{\Lambda} \cdot Q^T \\ = R^T \cdot R$$

$$\text{choose } R = \sqrt{\Lambda} \cdot Q^T$$

We know that $A = QR$

$$A^T A = (QR)^T QR = R^T \cdot Q^T \cdot Q \cdot R \\ = R^T \cdot I \cdot R \\ = R^T \cdot R = A$$

$$QR = A \\ R = Q^{-1} \cdot A \\ R = Q^T \cdot A$$

 $\Rightarrow A = R^T \cdot R \rightarrow R$ has independent columnsEven we can choose $R = Q \sqrt{\Lambda} Q^T$

Result The symmetric matrix A is positive definite if and only if there is a matrix R with independent columns such that $A = R^T \cdot R$

Example Test for positive^{definite} and positive semidefinite?

(23)

$$1) A = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & 1 \\ -1 & 1 & 2 \end{bmatrix} \xrightarrow{\substack{R_2 + \frac{1}{2}R_1 \\ R_3 + \frac{1}{2}R_1}} \begin{bmatrix} 2 & -1 & -1 \\ 0 & 3/2 & 1/2 \\ 0 & 1/2 & 3/2 \end{bmatrix}$$

$$2 - \frac{1}{2}$$

$$\frac{1}{2} - \frac{3}{2}x = 0$$

$$\frac{1}{2} - \frac{3}{2}x = 0$$

$$\frac{1}{2} = \frac{3}{2}x$$

$$x = \frac{1}{3}$$

$$\begin{bmatrix} 2 & -1 & -1 \\ 0 & 3/2 & 1/2 \\ 0 & 1/2 & 3/2 \end{bmatrix} \xrightarrow{R_3 - \frac{1}{3}R_2} \begin{bmatrix} 2 & -1 & -1 \\ 0 & 3/2 & 1/2 \\ 0 & 0 & 4/3 \end{bmatrix}$$

$$\frac{1}{2} - \frac{1}{3} \times \frac{3}{2}$$

$$\frac{3}{2} - \frac{1}{3} \times \frac{1}{2} = \frac{3}{2} - \frac{1}{6} = \frac{9-1}{6} = \frac{8}{6} = \frac{4}{3}$$

i) pivots are positive

$$ii) |A_1| = 2 > 0; A_2 = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}; |A_2| = 4 - 1 = 3 > 0$$

$$A_3 = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & 1 \\ -1 & 1 & 2 \end{bmatrix} \quad |A_3| = 2(4 - 1) + 1(-2 + 1) - 1(-1 + 2) \\ = 2(3) + (-1) - 1(1) \\ = 6 - 1 - 1 = 4 > 0$$

∴ All upper left submatrices have positive determinants.

iii) Eigen Values

$$\begin{vmatrix} 2-\lambda & -1 & -1 \\ -1 & 2-\lambda & 1 \\ -1 & 1 & 2-\lambda \end{vmatrix} = 0$$

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$$\begin{vmatrix} 2-\lambda & -1 & -1 \\ -1 & 2-\lambda & 1 \\ -1 & 1 & 2-\lambda \end{vmatrix} = 0$$

$$(2-\lambda) \{ (2-\lambda)^2 - 1 \} + 1 \{ -(2-\lambda) + 1 \} - 1 \{ -1 + (2-\lambda) \} = 0$$

$$(2-\lambda) \{ 4 + \lambda^2 - 4\lambda - 1 \} + \{ -2 + \lambda + 1 \} - 1 \{ -1 + 2 - \lambda \} = 0$$

$$(2-\lambda) \{ \lambda^2 - 4\lambda + 3 \} + \{ \lambda - 1 \} - \{ 1 - \lambda \} = 0$$

$$2\lambda^2 - 8\lambda + 6 - \lambda^3 + 4\lambda^2 - 3\lambda + \lambda - 1 - 1 + \lambda = 0$$

$$-\lambda^3 + 6\lambda^2 - 9\lambda + 4 = 0$$

$$\lambda^3 - 6\lambda^2 + 9\lambda - 4 = 0$$

$$1 - 6 + 9 - 4 = 0$$

$$\lambda = 1 \quad \begin{array}{ccc|c} 1 & -6 & 9 & -4 \\ 0 & 1 & -5 & 4 \\ \hline 1 & -5 & 4 & \boxed{0} \end{array}$$

$$\lambda^2 - 5\lambda + 4 = 0$$

$$\lambda^2 - 4\lambda - \lambda + 4 = 0$$

$$\lambda(\lambda - 4) - 1(\lambda - 4) = 0$$

$$(\lambda - 4)(\lambda - 1) = 0$$

$$\lambda = 1 \text{ and } \lambda = 4$$

∴ eigen values are 1, 1, 4 > 0

$$\text{iv) } x^T A x = \begin{bmatrix} x_1 & x_2 & x_3 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & 1 \\ -1 & 1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$= \begin{bmatrix} x_1 & x_2 & x_3 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 2x_1 - x_2 - x_3 \\ -x_1 + 2x_2 + x_3 \\ -x_1 + x_2 + 2x_3 \end{bmatrix}$$

$$x^T A x = x_1(2x_1 - x_2 - x_3) + x_2(-x_1 + 2x_2 + x_3) \\ + x_3(-x_1 + x_2 + 2x_3)$$

$$x^T A x = 2x_1^2 - x_1x_2 - x_1x_3 - x_1x_2 + 2x_2^2 + x_2x_3 \\ + x_2x_3 - x_1x_3 + 2x_3^2$$

$$\Rightarrow x^T A x = 2x_1^2 + 2x_2^2 + 2x_3^2 - 2x_1x_2 - 2x_1x_3 + 2x_2x_3$$

$$A = LDL^T = \begin{bmatrix} 1 & 0 & 0 \\ -1/2 & 1 & 0 \\ -1/2 & 1/3 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3/2 & 0 \\ 0 & 0 & 4/3 \end{bmatrix} \begin{bmatrix} 2 & -1 & -1 \\ 0 & 3/2 & 1/2 \\ 0 & 0 & 4/3 \end{bmatrix}$$

consider $x^T A x = x^T \{LDL^T\} x = (x^T L) D (L^T x)$

$$(L^T x)^T = x^T L$$

$$L^T x = \begin{bmatrix} 1 & -1/2 & -1/2 \\ 0 & 1 & 1/3 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{pmatrix} x - y/2 - z/2 \\ y + z/3 \\ z \end{pmatrix}$$

o.o $x^T A x = (x^T L) D (L^T x)$

$$= \begin{bmatrix} x - y/2 + z/2 & y + z/3 & z \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3/2 & 0 \\ 0 & 0 & 4/3 \end{bmatrix} \begin{bmatrix} x - y/2 + z/2 \\ y + z/3 \\ z \end{bmatrix}$$

$$= \begin{bmatrix} x - y/2 + z/2 & y + z/3 & z \end{bmatrix} \begin{bmatrix} 2(x - y/2 + z/2) \\ 3/2(y + z/3) \\ 4/3 z \end{bmatrix}$$

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$$x^T A x = \underbrace{2 \left(x - \frac{y}{2} + \frac{z}{2} \right)^2}_{>0} + \underbrace{\frac{3}{2} \left(y + \frac{z}{3} \right)^2}_{>0} + \underbrace{\frac{4}{3} z^2}_{>0}$$

$$\therefore x^T A x > 0$$

$\Rightarrow A$ is positive definite

Example check for positive definiteness?

$$A = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} \xrightarrow[R_3 + \frac{1}{2}R_1]{R_2 + \frac{1}{2}R_1} \begin{bmatrix} 2 & -1 & -1 \\ 0 & 3/2 & -3/2 \\ 0 & -3/2 & 3/2 \end{bmatrix} \xrightarrow{R_3 + R_2} \begin{bmatrix} 2 & -1 & -1 \\ 0 & 3/2 & -3/2 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{aligned} i) \quad x^T A x &= x^T \cdot L D L^T x = (L^T x)^T \cdot D \cdot (L^T x) \\ &= (x^T L) D (L^T x) \end{aligned}$$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ -1/2 & 1 & 0 \\ -1/2 & -1 & 0 \end{bmatrix} \quad L^T = \begin{bmatrix} 1 & -1/2 & -1/2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$L^T x = \begin{bmatrix} 1 & -1/2 & -1/2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$= \begin{pmatrix} x_1 - x_2/2 - x_3/2 \\ x_2 - x_3 \\ 0 \end{pmatrix}$$

$$(L^T x)^T = x_1 - \frac{x_2}{2} - \frac{x_3}{2} \quad \frac{x_2}{2} - \frac{x_3}{2} \quad 0$$

$$x^T A x = (L^T x)^T \cdot D \cdot L^T x$$

$$= \begin{pmatrix} x_1 - \frac{x_2}{2} - \frac{x_3}{2} & \frac{x_2}{2} - \frac{x_3}{2} & 0 \end{pmatrix} \begin{pmatrix} 2 & 0 & 0 \\ 0 & 3/2 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 - \frac{x_2}{2} - \frac{x_3}{2} \\ \frac{x_2}{2} - \frac{x_3}{2} \\ 0 \end{pmatrix}$$

$$x^T A x = 2 \left(x_1 - \frac{x_2}{2} - \frac{x_3}{2} \right)^2 + \frac{3}{2} \left(\frac{x_2}{2} - \frac{x_3}{2} \right)^2 \geq 0$$

$$\text{Pivots} = 2 > 0 \text{ \& } 3/2 > 0$$

$$A_1 = 2; |A_1| > 0; A_2 = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}; |A_2| = 4 - 1 = 3 > 0$$

$$A_3 = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}; |A_3| = 2(4-1) + 1(-2-1) - 1(1+2) \\ = 2(3) + 1(-3) - 1(3) \\ = 0$$

$$A = L D L^T = \underbrace{L \sqrt{D}}_{R^T} \cdot \underbrace{\sqrt{D} \cdot L^T}_R; \quad R = \sqrt{D} \cdot L^T \\ R^T = L \sqrt{D}$$

$$R = \begin{pmatrix} \sqrt{2} & 0 & 0 \\ 0 & \frac{\sqrt{3}}{\sqrt{2}} & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & -1/2 & -1/2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} \sqrt{2} & -1/\sqrt{2} & -1/\sqrt{2} \\ 0 & \frac{\sqrt{3}}{\sqrt{2}} & -\frac{\sqrt{3}}{\sqrt{2}} \\ 0 & 0 & 0 \end{pmatrix}$$

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If A is a symmetric matrix, then it is said to be positive semi-definite if :-

- 1) $x^T A x \geq 0 \quad \forall x$
- 2) All eigen values of A are greater than or equal to zero i.e $\lambda_i \geq 0$.
- 3) None of the principal submatrices have negative determinants.
- 4) Pivots are non-negative (≥ 0).
- 5) There exists a matrix R possibly with dependent columns such that $A = R^T \cdot R$

Example Test the matrix for positive semi-definiteness? $A = \begin{bmatrix} 5 & 2 & 1 \\ 2 & 2 & 2 \\ 1 & 2 & 5 \end{bmatrix}$

Solution :- We can verify that eigen values of A are $\lambda = 4, 4+2\sqrt{3}$ and $4-2\sqrt{3} > 0$.

$$\begin{bmatrix} 5 & 2 & 1 \\ 2 & 2 & 2 \\ 1 & 2 & 5 \end{bmatrix} \xrightarrow[R_3 - 1/5 R_1]{R_2 - 2/5 R_1} \begin{bmatrix} 5 & 2 & 1 \\ 0 & 6/5 & 8/5 \\ 0 & 8/5 & 24/5 \end{bmatrix} \xrightarrow[R_3 - \frac{4}{3} R_2]{R_3 - \frac{4}{3} R_2} \begin{bmatrix} 5 & 2 & 1 \\ 0 & 6/5 & 8/5 \\ 0 & 0 & 8/3 \end{bmatrix}$$

$$2 - 5x = 0 \quad 1 - 5y = 0$$

$$2 = 5x \quad 1 = 5y$$

$$x = 2/5 \quad y = 1/5$$

$$2 - \frac{2}{5}$$

$$\frac{8}{5} - \frac{6}{5}x = 0$$

$$\frac{5}{1} - \frac{1}{5}$$

$$\frac{8}{5} = \frac{6}{5}x$$

$$2 - \frac{2(2)}{5}$$

$$2 - \frac{2}{5}$$

$$x = 4/3$$

$$\frac{24}{5} - \frac{4}{3} \times \frac{8}{5}$$

$$\frac{72 - 32}{15} = \frac{40}{15} = \frac{8}{3}$$

1) The pivots are positive

$$2) |A_1| = 2 ; |A_2| = \begin{vmatrix} 5 & 2 \\ 2 & 2 \end{vmatrix} = 10 - 4 = 6$$

$$|A_3| = \begin{vmatrix} 5 & 2 & 1 \\ 2 & 2 & 2 \\ 1 & 2 & 5 \end{vmatrix} ; \begin{aligned} & 5(10-4) - 2(10-2) + 1(4-2) \\ & 5(6) - 2(8) + 2 \\ & = 30 - 16 + 2 = 16 > 0 \end{aligned}$$

The given matrix is positive definite.

Lectures 8 to 9 - The Singular Value Decomposition of a Matrix

Singular Value Decomposition :-

→ Any m by n matrix A can be factored into

$$A = U \Sigma V^T$$

(orthogonal) (diagonal) (orthogonal)

→ The columns of U (m by m) are eigen vectors of $A \cdot A^T$ and the columns of V (n by n) are eigen vectors of $A^T \cdot A$. The σ singular values on the diagonal of Σ (m by n) are the square roots of the non-zero eigen values of both $A \cdot A^T$ and $A^T \cdot A$.

Note :- For positive definite matrices Σ is A .

→ U and V give orthonormal bases for all four fundamental subspaces :-

first ' σ ' columns of U :- column space of A .

last ' $m - \sigma$ ' columns of U :- left null space of A

first ' σ ' columns of V :- row space of A

last ' $n - \sigma$ ' columns of V :- nullspace of A

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Remark 1

Eigen Vectors of $A \cdot A^T$ and $A^T \cdot A$ must go into the columns of U and V .

$$A \cdot A^T = (U \Sigma V^T)(V \Sigma^T U^T) = U \Sigma \Sigma^T U^T$$

Similarly $A^T A = V \Sigma^T \Sigma V^T$

$$AV = U\Sigma$$

$$A \cdot v_j = \sigma_j \cdot u_j$$

$$u_j = \frac{A \cdot v_j}{\sigma_j} \rightarrow \text{eigen vector in } A^T A$$

$$v_j = \frac{u_j^T \cdot A}{\sigma_j}$$

Example Find the SVD of $A = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix}_{2 \times 3}$

solution:-

$$A \cdot A^T = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ 2 & 3 \\ 2 & -2 \end{bmatrix} = \begin{bmatrix} 17 & 8 \\ 8 & 17 \end{bmatrix}$$

find

TO $\wedge$ eigen values of $A \cdot A^T \Rightarrow |A \cdot A^T - \lambda I| = 0$

$$\begin{vmatrix} 17-\lambda & 8 \\ 8 & 17-\lambda \end{vmatrix} = 0$$

$$(17-\lambda)^2 - 64 = 0$$

$$289 + \lambda^2 - 34\lambda - 64 = 0$$

$$\lambda^2 - 34\lambda + 225 = 0$$

$$\lambda^2 - 25\lambda - 9\lambda + 225 = 0$$

$$\lambda(\lambda - 25) - 9(\lambda - 25) = 0$$

$$\lambda = 25 \text{ or } \lambda = 9$$

$$\lambda_1 = 25$$

$$\lambda_2 = 9$$

For $\lambda_1 = 25$, eigen vectors are :-

$$\begin{bmatrix} 17-25 & 8 \\ 8 & 17-25 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$-8x + 8y = 0$$

$$8x - 8y = 0$$

$$8x = 8y$$

$$\text{let } y = 1 \Rightarrow x = 1 \therefore x_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

For $\lambda_2 = 9$ $[A - \lambda I]x = 0$

$$(17-\lambda)x + 8y = 0$$

$$8x + (17-\lambda)y = 0$$

$$8x + 8y = 0$$

$$8x + 8y = 0$$

$$8x = -8y$$

$$\text{let } y = 1 \Rightarrow x = -1$$

$$\therefore x_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

$$\text{Normalized vectors :- } u_1 = \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix} ; u_2 = \begin{pmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}$$

The singular values of A are :- $\sigma_1 = \sqrt{\lambda_1} = \sqrt{25} = 5$

$$\sigma_2 = \sqrt{\lambda_2} = \sqrt{9} = 3$$

consider $A \cdot A^T$

$$v_1 = \frac{u_1^T \cdot A}{\sigma_1} = \frac{\begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix}}{5} = \frac{1}{5} \begin{bmatrix} 5/\sqrt{2} \\ 5/\sqrt{2} \\ 0 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix}$$

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$$v_2 = \frac{u_2^T \cdot A}{\sigma_2} = \frac{\begin{bmatrix} -1 & 1 \\ \sqrt{2} & \sqrt{2} \end{bmatrix} \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix}}{3} = \frac{1}{3} \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \\ -4/\sqrt{2} \end{bmatrix}$$

$$v_2 = \begin{bmatrix} \frac{-1}{3\sqrt{2}} \\ \frac{1}{3\sqrt{2}} \\ \frac{-4}{3\sqrt{2}} \end{bmatrix}$$

v_3 should be orthogonal to both v_1 and v_2

∴ let $v_3 = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ $v_1^T \cdot v_3 = 0$ and $v_2^T \cdot v_3 = 0$

∴ $\frac{x}{\sqrt{2}} + \frac{y}{\sqrt{2}} = 0$ and $\frac{-x}{3\sqrt{2}} + \frac{y}{3\sqrt{2}} - \frac{4z}{3\sqrt{2}} = 0$

$$\begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ \frac{-1}{3\sqrt{2}} & \frac{1}{3\sqrt{2}} & \frac{-4}{3\sqrt{2}} \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\frac{-1}{3\sqrt{2}} - \frac{1}{\sqrt{2}} x = 0$$

$$\frac{-1}{3\sqrt{2}} = \frac{1}{\sqrt{2}} x$$

$$\boxed{x = -1/3}$$

$$\frac{-1}{3\sqrt{2}} + \frac{1}{3} \times \frac{1}{\sqrt{2}}$$

$$\frac{1}{3\sqrt{2}} + \frac{1}{3\sqrt{2}}$$

$$\begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ \frac{-1}{3\sqrt{2}} & \frac{1}{3\sqrt{2}} & \frac{-4}{3\sqrt{2}} \end{bmatrix} \xrightarrow[R_2 + 1/3 R_1]{R_2 \times 3} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & \frac{2}{3\sqrt{2}} & \frac{-4}{3\sqrt{2}} \end{bmatrix}$$

$$\begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & \frac{2}{3\sqrt{2}} & \frac{-4}{3\sqrt{2}} \end{bmatrix} \xrightarrow{R_2 - \frac{2}{3}R_1} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & \frac{-4}{3\sqrt{2}} \end{bmatrix}$$

$$\frac{2}{3\sqrt{2}} - \frac{1}{\sqrt{2}} x = 0$$

$$\frac{2}{3\sqrt{2}} = \frac{1}{\sqrt{2}} x$$

$$x = 2/3$$

solving,
on we get

$$v_3 = \begin{pmatrix} 2/3 \\ -2/3 \\ -1/3 \end{pmatrix}$$

$$z = \frac{-4}{3\sqrt{2}} R$$

$$\text{of } K = \frac{\sqrt{2}}{4}$$

$$= \frac{-4}{3\sqrt{2}} \frac{\sqrt{2}}{4} = -\frac{1}{3}$$

$$\therefore A = U \Sigma V^T = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 5 & 0 & 0 \\ 0 & 3 & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ -\frac{1}{3\sqrt{2}} & \frac{1}{3\sqrt{2}} & \frac{-4}{3\sqrt{2}} \\ \frac{2}{3} & -\frac{2}{3} & -\frac{1}{3} \end{bmatrix}$$

Example : Find the SVD of $A = \begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 2 & 2 \end{bmatrix}_{3 \times 2}$

Solution $A^T A = \underbrace{\begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 2 & 2 \end{bmatrix}}_A \begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 2 & 2 \end{bmatrix}$

$$A^T A = \begin{bmatrix} 9 & -1 \\ -1 & 9 \end{bmatrix}_{2 \times 2}$$

To find the eigen values of $A^T A$:-

$$\begin{vmatrix} 9-\lambda & -1 \\ -1 & 9-\lambda \end{vmatrix} = 0 \quad (9-\lambda)^2 - 1 = 0$$

$$81 + \lambda^2 - 18\lambda - 1 = 0$$

$$\lambda^2 - 18\lambda + 80 = 0$$

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$$\lambda^2 - 18\lambda + 80 = 0$$

$$\lambda^2 - 10\lambda - 8\lambda + 80 = 0$$

$$\lambda(\lambda - 10) - 8(\lambda - 10) = 0$$

$$(\lambda - 10)(\lambda - 8) = 0$$

$$\lambda_1 = 10 ; \lambda_2 = 8$$

$$[A^T A - \lambda I]x = 0$$

$$(9 - \lambda)x - y = 0$$

$$-x + (9 - \lambda)y = 0$$

$$\text{for } \lambda_1 = 10$$

$$-x - y = 0$$

$$-x - y = 0$$

$$-x = y$$

$$x = -y$$

$$x_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$\text{for } \lambda_2 = 8$$

$$x - y = 0$$

$$-x + y = 0$$

$$x_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Normalization :-

$$v_1 = \begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix} ; v_2 = \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix} \Rightarrow V = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \end{bmatrix}$$

The singular values of A are :-

$$\sigma_1 = \sqrt{\lambda_1} = \sqrt{10} ; \sigma_2 = \sqrt{\lambda_2} = \sqrt{8}$$

The eigen values of $A^T A$ are 10, 8, 0

$$u_1 = \frac{A \cdot v_1}{\sigma_1} = \frac{\begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}}{\sqrt{10}} = \frac{1}{\sqrt{10}} \begin{bmatrix} 2/\sqrt{2} \\ -4/\sqrt{2} \\ 0 \end{bmatrix}$$

$$u_1 = \begin{pmatrix} 1/\sqrt{5} \\ -2/\sqrt{5} \\ 0 \end{pmatrix}$$

$$u_2 = \frac{A \cdot v_2}{\sigma_2} = \frac{\begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}}{\sqrt{8}} = \frac{1}{\sqrt{8}} \begin{bmatrix} 0 \\ 0 \\ 4/\sqrt{2} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \frac{4}{2\sqrt{2} \cdot \sqrt{2}} \end{bmatrix}$$

$$u_2 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

u_3 should be orthogonal to both u_1 and u_2

$$x - 2y = 0$$

$$z = 0$$

$$x = 2y$$

$$\text{let } y = 1 \Rightarrow x = 2$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} \quad u_3 = \begin{pmatrix} 2/\sqrt{5} \\ 1/\sqrt{5} \\ 0 \end{pmatrix}$$

$$\therefore A = U \Sigma V^T$$

$$= \begin{pmatrix} 1/\sqrt{5} & 0 & 2/\sqrt{5} \\ -2/\sqrt{5} & 0 & 1/\sqrt{5} \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} \sqrt{10} & 0 \\ 0 & \sqrt{8} \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{pmatrix}$$

Example Find the SVD of $A = \begin{bmatrix} 6 & -2 & -1 \\ -2 & 6 & -1 \\ -1 & -1 & 5 \end{bmatrix}$

solution - we observe that $A^T = A$

i) we can verify that the eigen values are positive.

$$|A_1| = 6; |A_2| = \begin{vmatrix} 6 & -2 \\ -2 & 6 \end{vmatrix} = 36 - 4 = 32 > 0$$

$$\begin{aligned} |A_3| &= 6\{30 - 1\} + 2\{-10 - 1\} - 1\{2 + 6\} \\ &= 6\{29\} + 2\{-11\} - 1\{8\} \\ &= 174 - 22 - 8 = 144 > 0 \end{aligned}$$

36. A is positive definite

$$A = G \wedge G^T$$

To find eigen values :-

$$\begin{vmatrix} 6-\lambda & -2 & -1 \\ -2 & 6-\lambda & -1 \\ -1 & -1 & 5-\lambda \end{vmatrix} = 0$$

$$(6-\lambda) \{ (6-\lambda)(5-\lambda) - 1 \} + 2 \{ -2(5-\lambda) - 1 \} - 1 \{ 2 + (6-\lambda) \} = 0$$

$$(6-\lambda) \{ 30 - 6\lambda - 5\lambda + \lambda^2 - 1 \} + 2 \{ -10 + 2\lambda - 1 \} - 1 \{ 8 - \lambda \} = 0$$

$$(6-\lambda) \{ \lambda^2 - 11\lambda + 29 \} - 20 + 4\lambda - 2 - 8 + \lambda = 0$$

$$6\lambda^2 - 66\lambda + 174 - \lambda^3 + 11\lambda^2 - 29\lambda - 30 + 5\lambda = 0$$

$$-\lambda^3 + 17\lambda^2 - 90\lambda + 144 = 0$$

$$\lambda^3 - 17\lambda^2 + 90\lambda - 144 = 0$$

$$27 - 153 + 270 - 144 = 0$$

$$\lambda = 3$$

$$\lambda = 3 \begin{array}{c|cccc} 1 & -17 & 90 & -144 \\ 0 & 3 & -42 & 144 \\ \hline 1 & -14 & 48 & 0 \end{array}$$

$$\lambda^2 - 14\lambda + 48 = 0$$

$$\lambda^2 - 8\lambda - 6\lambda + 48 = 0$$

$$\lambda(\lambda - 8) - 6(\lambda - 8) = 0$$

$$(\lambda - 8)(\lambda - 6) = 0$$

$$\lambda = 6 \text{ or } \lambda = 8$$

$$\lambda = 8, 6, 3$$

To find the eigen vectors :-

$$(6-\lambda)x - 2y - z = 0$$

$$-2x + (6-\lambda)y - z = 0$$

$$-x - y + (5-\lambda)z = 0$$

$$\text{for } \lambda = 8$$

$$-2x - 2y - z = 0$$

$$-2x - 2y - z = 0$$

$$-x - y - 3z = 0$$

$$\begin{bmatrix} -2 & -2 & -1 \\ -2 & -2 & -1 \\ -1 & -1 & -3 \end{bmatrix} \xrightarrow[R_3 - \frac{1}{2}R_1]{R_2 - R_1} \begin{bmatrix} -2 & -2 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & -5/2 \end{bmatrix}$$

$$\begin{aligned} -\frac{3}{2} + \frac{1}{2} \\ \frac{1}{2} \\ \frac{-6+1}{2} \end{aligned}$$

$$z = 0$$

$$-2x - 2y = 0$$

$$-2x = 2y$$

$$x = -y$$

$$\text{let } y = 1 \Rightarrow x = -1$$

$$\text{so } x_1 = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$$

for $\lambda = 6$

$$-2y - z = 0$$

$$-2x - z = 0$$

$$-x - y - z = 0$$

$$\begin{bmatrix} 0 & -2 & -1 \\ -2 & 0 & -1 \\ -1 & -1 & -1 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_1} \begin{bmatrix} -2 & 0 & -1 \\ 0 & -2 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

$$\begin{bmatrix} -2 & 0 & -1 \\ 0 & -2 & -1 \\ -1 & -1 & -1 \end{bmatrix} \xrightarrow[R_3 - \frac{1}{2}R_1]{R_3 \times \frac{1}{2}} \begin{bmatrix} -2 & 0 & -1 \\ 0 & -2 & -1 \\ 0 & -1 & -1/2 \end{bmatrix} \xrightarrow[R_3 - \frac{1}{2}R_2]{R_3 \times \frac{1}{2}} \begin{bmatrix} -2 & 0 & -1 \\ 0 & -2 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$-2x - z = 0$$

$$-2x = z$$

$$x = -z/2$$

$$x = -1$$

$$x_2 = \begin{pmatrix} -1 \\ -1 \\ 2 \end{pmatrix}$$

$$-1/2 + 1/2$$

$$-2y - z = 0$$

$$-2y = +z$$

$$y = -z/2$$

$$\text{let } z = 1$$

$$\text{so } y = -1/2$$

$$\text{If } z = 2 \Rightarrow y = -1$$

$$\begin{aligned} -1 + \frac{2}{3} \\ \frac{1}{3} \\ \frac{-3+2}{3} \end{aligned}$$

for $\lambda = 3$

$$3x - 2y - z = 0$$

$$-2x + 3y - z = 0$$

$$-x - y + 2z = 0$$

$$\begin{bmatrix} 3 & -2 & -1 \\ -2 & 3 & -1 \\ -1 & -1 & 2 \end{bmatrix} \xrightarrow{R_2 + \frac{2}{3}R_1} \begin{bmatrix} 3 & -2 & -1 \\ 0 & 5/3 & -5/3 \\ -1 & -1 & 2 \end{bmatrix}$$

$$-2 - 3x = 0$$

$$-2 = 3x \Rightarrow x = -2/3$$

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$$\begin{bmatrix} 3 & -2 & -1 \\ 0 & 5/3 & -5/3 \\ -1 & -1 & 2 \end{bmatrix} \xrightarrow{R_3 + 1/3 R_1} \begin{bmatrix} 3 & -2 & -1 \\ 0 & 5/3 & -5/3 \\ 0 & -5/3 & 5/3 \end{bmatrix} \xrightarrow{R_3 + R_2} \begin{bmatrix} 3 & -2 & -1 \\ 0 & 5/3 & -5/3 \\ 0 & 0 & 0 \end{bmatrix}$$

$$-1 - 3x = 0$$

$$-1 = 3x$$

$$x = -1/3$$

$$\frac{-1 - 2}{1} = \frac{-3 - 2}{3}$$

$$\frac{2 - 1}{1} = \frac{1}{3}$$

$$\frac{5}{3}y = \frac{5}{3}z$$

$$y = z$$

$$\therefore x_3 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$3x - 2y - z = 0$$

$$3x - 2z - z = 0$$

$$3x - 3z = 0$$

$$x = 3z/3 = z$$

$$a_1 = \begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \\ 0 \end{pmatrix}; a_2 = \begin{pmatrix} -1/\sqrt{6} \\ -1/\sqrt{6} \\ 2/\sqrt{6} \end{pmatrix}; a_3 = \begin{pmatrix} 1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{pmatrix}$$

$$\therefore Q = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{6} & 1/\sqrt{3} \\ -1/\sqrt{2} & -1/\sqrt{6} & 1/\sqrt{3} \\ 0 & 2/\sqrt{6} & 1/\sqrt{3} \end{bmatrix}$$

$$\therefore \boxed{A = Q \wedge Q^T}$$

Pseudo-Inverse

→ The pseudo-inverse A^+ of an ' $m \times n$ ' matrix A is an extension of the inverse of a ^{square} matrix to non-square matrices and to singular (non-invertible) square matrices.

→ The pseudo-inverse matrix ' A^+ ' is an ' $n \times m$ ' matrix with the following properties :-

1) If $m \geq n$, then $A^T A$ is invertible and
 $A^+ = (A^T A)^{-1} \cdot A^T$ and so $A^+ A = I$, A^+ is a left inverse of A .

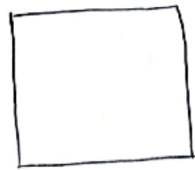
2) If $m \leq n$, then $A \cdot A^T$ is invertible and
 $A^+ = A^T (A \cdot A^T)^{-1}$ and so $A \cdot A^+ = I$, i.e., A^+ is a right inverse of A .

Least Squares :-

→ For a rectangular system ' $Ax = b$ ', the least squares solution comes from the normal equation :-

$$A^T \cdot A \cdot \hat{x} = A^T b$$

→ Here in solving $Ax = b$ has two possible dependent rows or dependent columns of A .



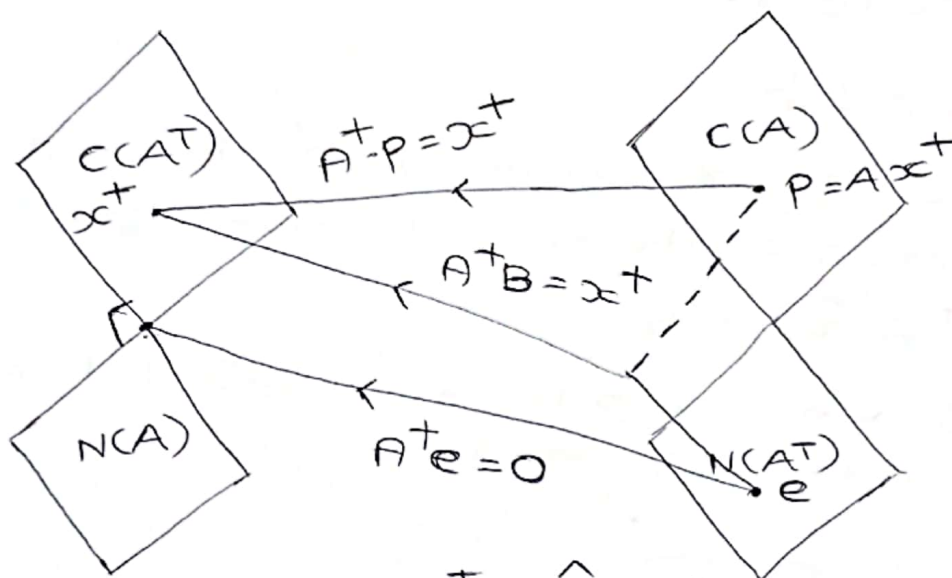
$C(A)$

→ Then instead of $Ax = b$, we can solve $A^T \cdot A \cdot \hat{x} = A^T b$.

→ If A has dependent columns, then $\hat{x}$ will not be unique. Then, we have to choose a particular solution of $A^T \cdot A \cdot \hat{x} = A^T b$ and we have to choose the shortest.

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- The optimal solution of ' $Ax=b$ ' is the minimum length x^+ solution of $A^T \cdot A \cdot \hat{x} = A^T \cdot b$.
 - Minimum length solution = x^+
 - x^+ is the best solution of $Ax=b$ (with no solution).
 - $x^+ = A^+ B$
 - ↳ Pseudo Inverse

→ Suppose $A = U \Sigma V^T$ is SVD of A . Then the pseudo inverse $A^+ = V \Sigma^+ U^T$ where Σ^+ is the diagonal matrix formed by taking the reciprocal of each non zero element in Σ , leaving the zeros in place and then transforming the resulting matrix.



x_0 also solves $A^T \cdot A \cdot \hat{x}_0 = A^T b$

→ row space component

Since $A \cdot x_n = 0$

→ null space component

All solutions of $A^T A \cdot \hat{x} = A^T b$ have same x_0 that vector is x^+ .

$$\Sigma = \begin{bmatrix} \sigma_1 & 0 & 0 & \dots & 0 \\ 0 & \sigma_2 & 0 & \dots & 0 \\ 0 & 0 & \sigma_3 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & \sigma_r \\ \vdots & \vdots & \vdots & \ddots & \vdots \end{bmatrix}$$

$m \times n$

$$\Sigma^+ = \begin{bmatrix} \frac{1}{\sigma_1} & 0 & 0 & \dots & 0 \\ 0 & \frac{1}{\sigma_2} & 0 & \dots & 0 \\ 0 & 0 & \frac{1}{\sigma_3} & \dots & 0 \\ 0 & 0 & 0 & \frac{1}{\sigma_4} & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & \dots & \frac{1}{\sigma_r} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \end{bmatrix}$$

$n \times m$

Example using SVD to find least squares solution of the system :- $x_2 + 2x_3 = 5$; $x_1 + x_3 = 1$

solution :- $A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 1 \end{bmatrix}$; $b = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$

using SVD $A = U \Sigma V^T$

we can verify that :- $U = \begin{bmatrix} 2/\sqrt{5} & -1/\sqrt{5} \\ 1/\sqrt{5} & 2/\sqrt{5} \end{bmatrix}$

$$\Sigma = \begin{bmatrix} \sqrt{5} & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

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$$V = \begin{bmatrix} \frac{1}{\sqrt{30}} & \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \\ \frac{2}{\sqrt{30}} & \frac{-1}{\sqrt{5}} & \frac{2}{\sqrt{6}} \\ \frac{5}{\sqrt{30}} & 0 & \frac{-1}{\sqrt{6}} \end{bmatrix}$$

Least Square solution :- $x^+ = A^+ b$

we have $A^+ = V \Sigma^+ U^T$

$$x^+ = \underbrace{V \Sigma^+ U^T}_{3 \times 2} b$$

$$x^+ = \begin{bmatrix} \frac{1}{\sqrt{30}} & \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \\ \frac{2}{\sqrt{30}} & \frac{-1}{\sqrt{5}} & \frac{2}{\sqrt{6}} \\ \frac{5}{\sqrt{30}} & 0 & \frac{-1}{\sqrt{6}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{6}} & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \\ \frac{-1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \end{bmatrix} \begin{bmatrix} 5 \\ 1 \end{bmatrix}$$

3×2 2×2

$$= \begin{bmatrix} \frac{1}{\sqrt{30}} & \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \\ \frac{2}{\sqrt{30}} & \frac{-1}{\sqrt{5}} & \frac{2}{\sqrt{6}} \\ \frac{5}{\sqrt{30}} & 0 & \frac{-1}{\sqrt{6}} \end{bmatrix} \begin{bmatrix} \frac{2}{\sqrt{30}} & \frac{1}{\sqrt{30}} \\ \frac{-1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 5 \\ 1 \end{bmatrix}$$

3×3 3×2

$$= \begin{bmatrix} \frac{2}{30} - \frac{1}{5} & \frac{1}{30} + \frac{4}{5} \\ \frac{4}{30} + \frac{1}{5} & \frac{2}{30} - \frac{2}{5} \\ \frac{10}{30} & \frac{5}{30} \end{bmatrix} \begin{bmatrix} 5 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{2-6}{30} & \frac{1+24}{30} \\ \frac{4+6}{30} & \frac{2-12}{30} \\ \frac{1}{3} & \frac{1}{6} \end{bmatrix}$$

$$x^+ = \begin{bmatrix} -\frac{4}{30} & \frac{25}{30} \\ \frac{1}{3} & -\frac{1}{3} \\ \frac{1}{3} & \frac{1}{6} \end{bmatrix} \begin{bmatrix} 5 \\ 1 \end{bmatrix} \quad \begin{matrix} 3 \times 2 \\ 2 \times 1 \end{matrix} = \begin{bmatrix} -\frac{20}{30} + \frac{25}{30} \\ \frac{5}{3} - \frac{1}{3} \\ \frac{5}{3} + \frac{1}{6} \end{bmatrix} = \begin{bmatrix} \frac{1}{6} \\ \frac{4}{3} \\ \frac{10+1}{6} \end{bmatrix}$$

$$x^+ = \begin{bmatrix} \frac{1}{6} \\ \frac{4}{3} \\ \frac{11}{6} \end{bmatrix}$$

Lecture 11 - Applications of SVD

→ Covariance of a matrix [2x2]

→ Example on constructing co-variance matrix :-

We can represent the covariance matrix by its eigen vectors and eigen values :-

$$\Sigma \vec{v} = \lambda \cdot \vec{v}$$

where $\vec{v}$ is an eigen vector of Σ and λ is the corresponding eigen value.

→ Equation above holds for each eigen vector - eigen value pair of matrix Σ . In the 2D case, we obtain two eigen values and two eigen vectors. The system of two equations defined by equation represented efficiently using matrix notation :-

$$\boxed{\Sigma V = V L}$$

→ This means that we can represent the covariance matrix as a function of its eigenvectors and eigenvalues:

$$\Sigma = V L V^{-1}$$

→ This is called the eigendecomposition of the CO-Variance matrix and can be obtained using a Singular Value Decomposition algorithm.

→ Whereas the eigenvectors represent the directions of the largest variance of the data, the eigen values represents the magnitude of this variance in those directions. In other words, V represents a rotation matrix, while L represents a scaling matrix.

→ The CO-Variance matrix can thus be decomposed further as :-

$$\Sigma = R S S R^{-1}$$

→ where $R = V$ is a rotation matrix and $S = \sqrt{L}$ is a scaling matrix.

→ Linear transformation $T = RS$

→ Since S is a diagonal scaling matrix, $S = S^T$.

→ Furthermore, R is an orthogonal matrix, $R^{-1} = R^T$.

→ $T^T = (RS)^T = S^T \cdot R^T = S \cdot R^{-1}$

→ The covariance matrix can thus be written as :-

$$\Sigma = RS \cdot S R^{-1} = T \cdot T^T$$

→ In other words, if we apply the linear transformation defined by $T = RS$ to the original data D , we obtain the rotated and scaled data.

→ D' with covariance matrix $\circ T \cdot T^T = \Sigma' = R \cdot S \cdot R^{-1}$

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→ The covariance matrix represents a linear transformation of the original data.

→ The largest eigen vector, i.e. the eigen vector with the corresponding eigen value, always points in the direction of the largest variance of the data and thereby defines its orientation.

→ Subsequent eigen vectors are always orthogonal to the largest eigen vector due to the orthogonality of rotation matrices.

Example Find the co-variance matrix given that $\circ$ -

$$X_1 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}; X_2 = \begin{bmatrix} 4 \\ 2 \\ 13 \end{bmatrix}; X_3 = \begin{bmatrix} 7 \\ 8 \\ 1 \end{bmatrix}; X_4 = \begin{bmatrix} 8 \\ 4 \\ 5 \end{bmatrix}$$

Solution $\circ$ Step 1 - Find the mean vector

$$\text{Mean} = \frac{1}{n} [X_1 + X_2 + X_3 + X_4] = \begin{bmatrix} 5 \\ 4 \\ 5 \end{bmatrix} = M$$

Step 2 The mean deviation of the vectors

$X_{oi}; i = 1, 2, 3, 4$ are $\circ$ -

$$\hat{X}_{oi} = X_{oi} - M, i = 1, 2, 3, 4$$

$$\hat{X}_1 = \begin{bmatrix} -4 \\ -2 \\ -4 \end{bmatrix}; \hat{X}_2 = \begin{bmatrix} -1 \\ -2 \\ 8 \end{bmatrix}; \hat{X}_3 = \begin{bmatrix} 2 \\ 4 \\ -4 \end{bmatrix}; \hat{X}_4 = \begin{bmatrix} 3 \\ 0 \\ 0 \end{bmatrix}$$

Let the matrix $B = [\hat{X}_1 \hat{X}_2 \hat{X}_3 \hat{X}_4]$

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$$B = \begin{bmatrix} -4 & -1 & 2 & 3 \\ -2 & -2 & 4 & 0 \\ -4 & 8 & -4 & 0 \end{bmatrix}$$

The covariance matrix is $S = \frac{1}{n-1} [B \cdot B^T]$

$$S = \frac{1}{3} \begin{bmatrix} -4 & -1 & 2 & 3 \\ -2 & -2 & 4 & 0 \\ -4 & 8 & -4 & 0 \end{bmatrix}_{3 \times 4} \begin{bmatrix} -4 & -2 & -4 \\ -1 & -2 & 8 \\ 2 & 4 & -4 \\ 3 & 0 & 0 \end{bmatrix}_{4 \times 3} = \frac{1}{3} \begin{bmatrix} 30 & 18 & 0 \\ 18 & 24 & -24 \\ 0 & -24 & 96 \end{bmatrix}$$

$$S = \begin{bmatrix} 10 & 6 & 0 \\ 6 & 8 & -8 \\ 0 & -8 & 32 \end{bmatrix}$$

Example The table below lists the weights and heights of 5 boys. Find the covariance matrix :-

Boy :	1	2	3	4	5
Weight :	120	125	125	135	150
Height :	61	60	64	68	72

Solution :- Let $X = \begin{pmatrix} \text{Weight} \\ \text{Height} \end{pmatrix}$

$$\text{Mean of } X \text{ is } M = \frac{1}{5} [X_1 + X_2 + X_3 + X_4 + X_5]$$

$$M = \frac{1}{5} \begin{bmatrix} 655 \\ 325 \end{bmatrix} = \begin{bmatrix} 131 \\ 65 \end{bmatrix}$$

$$\text{Let } \hat{x}_i = x_i - M, \quad i = 1, 2, 3, 4, 5$$

$$\hat{x}_1 = \begin{bmatrix} -11 \\ -4 \end{bmatrix}; \hat{x}_2 = \begin{bmatrix} -6 \\ -5 \end{bmatrix}; \hat{x}_3 = \begin{bmatrix} -6 \\ -1 \end{bmatrix}; \hat{x}_4 = \begin{bmatrix} 4 \\ 3 \end{bmatrix}; \hat{x}_5 = \begin{bmatrix} 19 \\ 7 \end{bmatrix}$$

let

$$B = [\hat{x}_1 \hat{x}_2 \hat{x}_3 \hat{x}_4 \hat{x}_5]$$

$$B = \begin{bmatrix} -11 & -6 & -6 & 4 & 19 \\ -4 & -5 & -1 & 3 & 7 \end{bmatrix}_{2 \times 5}$$

$$S = \frac{1}{n-1} [B \cdot B^T]$$

$$S = \frac{1}{4} \begin{bmatrix} -11 & -6 & -6 & 4 & 19 \\ -4 & -5 & -1 & 3 & 7 \end{bmatrix} \begin{bmatrix} -11 & -4 \\ -6 & -5 \\ -6 & -1 \\ 4 & 3 \\ 19 & 7 \end{bmatrix}$$

$$S = \frac{1}{4} \begin{bmatrix} 570 & 225 \\ 225 & 100 \end{bmatrix} = \begin{bmatrix} 142.5 & 56.25 \\ 56.25 & 25 \end{bmatrix}$$